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Subject Code:BP302T

Roll No:

BPHARM
(SEM. III) THEORY EXAMINATION 2025-26
PHYSICAL PHARMACEUTICS I

TIME: 3 HRS**M.MARKS: 75**

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 2 = 20**

a.	Define dielectric constant and state its significance.
b.	What is critical micelle concentration (CMC)?
c.	Define partition coefficient.
d.	What is zeta potential?
e.	Define adsorption isotherm.
f.	What is Nernst distribution law?
g.	Define refractive index.
h.	What is eutectic mixture?
i.	Define sedimentation volume.
j.	What is Brownian movement?

SECTION B

2. Attempt any two parts of the following:**2 x 10 = 20**

a.	Explain Nernst Distribution Law. Discuss its assumptions, limitations, and pharmaceutical applications.
b.	Describe electrical properties of colloidal dispersions. Explain zeta potential, electrical double layer, and methods of determination.
c.	Discuss the kinetics of drug degradation. Explain zero-order, first-order, and pseudo-first-order reactions with pharmaceutical examples.

SECTION C

3. Attempt any five parts of the following:**7 x 5 = 35**

a.	Discuss phase rule and explain one-component and two-component systems with suitable diagrams.
b.	Explain refractive index. Describe the principle, construction, and working of Abbé refractometer.
c.	Discuss colloidal dispersions. Classify colloids and explain their pharmaceutical applications.
d.	Explain sedimentation and diffusion. Discuss factors affecting sedimentation rate.
e.	Describe eutectic systems and their pharmaceutical significance with suitable examples.
f.	Explain dielectric constant and dipole moment. Discuss their importance in pharmacy.
g.	Explain polymorphism and its pharmaceutical significance.